Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/19/2024
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Multi-Environment Virtual Training for Long Duration Exploration Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-5004	**Congressional District:**	2
Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	05/19/2021	**End Date:**	05/18/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder) Robinson, Stephen Ph.D. (University of California, Davis) Vance, Eric Ph.D. (University of Colorado, Boulder) Szafir, Daniel (University of North Carolina)		
Grant/Contract No.:	80NSSC21K1140		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of adaptive VR for training in long duration exploration missions (LDEM). These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry decent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (EVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth prior to the start of the mission, and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will experimentally investigate how skills transfer from the training environment to a physical setting, the degree to which the training environment enables skills maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, we will investigate training in an adaptive, complex VR environment as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. The novel contributions of this research are to investigate the use of immersive, adaptive VR environments to train astronauts specifically designed for LDEMs by 1) Incorporating multiple, mission critical tasks for a mission to Mars; 2) Investigating skills transfer, retention, and generalizability; 3) Designing with low volume, mass, and operational overhead; and 4) Investigating the degree to which the proposed training can also provide secondary benefits of broad brain stimulation. Our proposed research is directly aligned with that of the Virtual NASA Specialized Center of Research (VNSCOR), and can be integrated with researchers investigating teams, optimal adaptive control algorithms, motivational strategies, or neural and cognitive decrements with both human and animal studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has wide application in training in immersive environments, particularly for complex tasks. We develop a novel training algorithm to facilitate rapid skills acquisition, while also ensuring broad improvement in performance (rather than unidimensional performance). To our knowledge is absent from the literature, but may improve how we administer training in an automated (rather than supervised) training paradigm. This research will also contribute to our understanding of immersive environments and the degree to which they stimulate the brain. This informs our scientific understanding of neuroplasticity when learning and acquiring new skills. This has applications in rehabilitation, to inform how using these environments may yield better patient outcomes.
Task Progress:	Historically, NASA has led the way in virtual reality (VR) technology development to train astronauts for microgravity environments, but there remain open questions regarding the use of VR for training in long duration exploration (LDEM) missions. These missions are likely to have long time periods between pre-mission training and execution of a mission task. Crews will need to train and maintain skills on tasks critical to mission success, such as entry descent and landing (EDL), habitat maintenance and repair, and planetary surface extravehicular activity (sEVA). To be successful, performance on these tasks will need to be maintained after high-fidelity training on Earth pre-mission and refreshed during transit prior to execution on the surface. The objective of this research is to develop an effective training environment for mission critical LDEM tasks. We will investigate how skills transfer from the VR training environment to a physical setting, the degree to which the training environment enables performance maintenance over longer durations, and the generalizability of skills learned in the training environment when transferring to un-trained tasks. As a secondary benefit, training in an adaptive, complex environment could serve as a countermeasure to the neural decrements demonstrated in spaceflight. Thus, this research will quantify neurophysiological response in the cerebral cortex during training to assess the potential benefit of the trainer as a countermeasure. In Year 3, we used our EDL virtual reality simulator to complete four novel and separate human experimentation investigations related to algorithm validation, self-led learning, ecological validity, and skill retention. We further developed our sEVA virtual simulator and physical mockup to be used in another validation investigation and a neuroimaging investigation to evaluate the neural effects of virtual reality operations compared to representative corollary tasks. Finally, further development of the habitat virtual simulator has been made in hopes of validation testing this summer. Progress toward using our EDL simulator aboard the Polaris Dawn crew mission has also been made. In future years, Trinity will finalize data collection and analysis on neural activation during VR training within the sEVA environment, complete the development of the habitat environment, further investigate the effects of continuous training in complex vs. simple (corollary) VR environments on skill retention over long periods, collect and analyze data from the Polaris Dawn mission, and apply the VR training environments developed for use by an analog crew during a Human Exploration Research Analog (HERA) campaign.
Bibliography Type:	Description: (Last Updated: 02/18/2026)
Abstracts for Journals and Proceedings	Bonarrigo LE, Putman E, Rees W, Peterson B, Li I, Sherman SO, Clark TK, Szafir DJ, Anderson AP. "Quantifying neural activation during spaceflight-relevant virtual reality training." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Bonarrigo LE, Putman E, Rees W, Peterson B, Galvin E, Verniani A, Tredinnick S, Sherman S, Baldonado K, Vance E, Robinson S, Szafir D, Clark TK, Anderson AP. "Development and assessment of a virtual reality trainer for long duration spaceflight." Aerospace Medical Association (AsMA) 2024 Annual Scientific Meeting, Chicago, Illinois, May 5-10, 2024. Abstracts. Aerospace Medical Association (AsMA) 2024 Annual Scientific Meeting, Chicago, Illinois, May 5-10, 2024. , May-2024
Abstracts for Journals and Proceedings	Verniani A, Galvin E, Tredinnick S, Vance E, Anderson AP. "Evaluating astronaut training algorithms in virtual reality for long-duration exploration missions." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023

Articles in Peer-reviewed Journals	Verniani A, Galvin E, Tredinnick S, Putman E, Vance E, Clark TK, Anderson AP. "Features of adaptive training algorithms for improved complex skill acquisition." Front Virtual Real. 2024 Feb 8;5:1322656. https://doi.org/10.3389/frvir.2024.1322656 , Feb-2024